

Class 9 Science – Motion (Annual Exam 2026)

20 Highly Probable Questions with Full Exam-Ready Answers

Conceptual & Theoretical Questions

1. Distinguish between Distance and Displacement. When is displacement equal to distance?

Answer:

Distance is the **total length of the actual path** travelled by an object, irrespective of direction. It is always positive.

Displacement is the **shortest straight-line distance** between the initial and final positions of an object and has a definite direction.

The magnitude of displacement is equal to the distance travelled **only when the object moves along a straight line without changing direction**.

Keywords: actual path, shortest distance, direction, scalar, vector, straight-line motion

2. Define scalar and vector quantities with two examples each from this chapter.

Answer:

Scalar quantities are physical quantities that have **magnitude only**. Examples from this chapter include **distance** and **speed**.

Vector quantities are physical quantities that have **both magnitude and direction**. Examples include **displacement** and **velocity**.

Keywords: magnitude only, magnitude and direction, distance, speed, displacement, velocity

3. Explain uniform and non-uniform motion with one example each.

Answer:

Uniform motion is the motion in which an object covers **equal distances in equal intervals of time**.

Example: A car moving at constant speed on a straight road.

Non-uniform motion is the motion in which an object covers **unequal distances in equal intervals of time**.

Example: A freely falling stone.

Keywords: equal distances, unequal distances, constant speed, changing speed

4. Define acceleration. State its SI unit. When is acceleration positive or negative?

Answer:

Acceleration is defined as the **rate of change of velocity with time**.

Its SI unit is **m s^{-2}** .

Acceleration is **positive** when velocity increases with time and **negative (retardation)** when velocity decreases with time.

Keywords: rate of change of velocity, m s^{-2} , increase in velocity, decrease in velocity

5. Define average speed and average velocity. Can average velocity be zero?

Answer:

Average speed is the **total distance travelled divided by total time taken**.

Average velocity is the **total displacement divided by total time taken**.

Yes, average velocity can be zero when the displacement is zero, such as when an athlete completes one full round of a circular track and returns to the starting point.

Keywords: total distance, total displacement, circular path, zero displacement

6. What does an odometer measure? How is it different from a speedometer?

Answer:

An odometer measures the **total distance travelled** by a vehicle.

A speedometer measures the **instantaneous speed** of a vehicle at a given moment.

Keywords: distance travelled, instantaneous speed, automobile

7. Why is uniform circular motion called accelerated motion?

Answer:

In uniform circular motion, although the **speed remains constant**, the **direction of velocity changes continuously**. Since velocity changes, the object is said to be in accelerated motion.

Keywords: change in direction, velocity change, centripetal acceleration

8. Derive the three equations of motion using graphical method.

Answer:

Consider a velocity–time graph for an object moving with uniform acceleration.

1. First equation:

Slope of v–t graph = acceleration

$$a = \frac{v - u}{t} \Rightarrow v = u + at$$

2. Second equation:

Distance travelled = area under v–t graph

$$s = ut + \frac{1}{2}at^2$$

3. Third equation:

Eliminating time t from first two equations:

$$v^2 - u^2 = 2as$$

Keywords: velocity-time graph, slope, area under graph, uniform acceleration

9. Can an object have zero velocity but non-zero acceleration? Give an example.

Answer:

Yes. At the **highest point of a vertically upward thrown ball**, its velocity becomes zero, but acceleration due to gravity (g) still acts downward.

Keywords: highest point, gravity, vertical motion

Graph-Based Questions

10. What does the slope of a distance-time and velocity-time graph represent?

Answer:

The slope of a **distance-time graph** represents the **speed** of the object.

The slope of a **velocity-time graph** represents the **acceleration** of the object.

Keywords: slope, speed, acceleration

11. What does the area under a velocity-time graph represent?

Answer:

The area under a velocity-time graph represents the **displacement** of the object.
Its SI unit is **metre (m)**.

Keywords: area, displacement, metre

12. Describe the distance-time graph for an object at rest and with non-uniform speed.

Answer:

For an object at rest, the distance-time graph is a **horizontal straight line** parallel to the time axis.
For non-uniform speed, the distance-time graph is a **curved line**.

Keywords: horizontal line, curved line, rest, non-uniform motion

13. What does a velocity-time graph parallel to the time axis indicate?

Answer:

A velocity-time graph parallel to the time axis indicates that the object is moving with **uniform velocity** and **zero acceleration**.

Keywords: constant velocity, zero acceleration

Numerical & Applied Questions

14. Calculate average speed for a journey with different speeds.

Answer:

Average speed is calculated using:

$$\text{Average speed} = \frac{\text{Total distance}}{\text{Total time}}$$

(Time for each part is calculated separately and added.)

Keywords: total distance, total time

15. A bus starts from rest and accelerates uniformly. Find final speed and distance.

Answer:

Given: $u = 0$, a and t are known

Using: $v = u + at$ and $s = ut + \frac{1}{2}at^2$

Keywords: equations of motion, uniform acceleration

16. Find the stopping distance of a train under uniform retardation.

Answer:

Using third equation of motion:

$$v^2 - u^2 = 2as$$

Where final velocity $v = 0$.

Keywords: retardation, stopping distance

17. Find the speed of a satellite in circular orbit.

Answer:

Speed is given by:

$$v = \frac{2\pi r}{T}$$

Keywords: circular motion, orbital speed

18. Solve a problem involving acceleration, constant speed, and retardation.

Answer:

Motion is divided into parts and equations of motion are applied separately. Total distance is the **sum of distances** in all parts.

Keywords: multiple motion stages, total distance

19. Find maximum height and time for a vertically thrown stone.

Answer:

At maximum height, $v = 0$

Using: $v = u - gt$ and $v^2 = u^2 - 2gh$

Keywords: gravity, vertical motion

20. Find distance and displacement on a circular track.

Answer:

Distance = number of rounds $\times$ circumference

Displacement depends on initial and final position (zero for full rounds).

Keywords: circular track, circumference, zero displacement

 Exam Tip

Write **formula** $\rightarrow$ **substitution** $\rightarrow$ **final answer with unit** for full marks.